Joint PDF of Multiple Random Variables

Lecture 7

Topics in this lecture

- Joint Probability Density Function
- Joint Cumulative Distribution Function
- Conditional PDF
- Conditional Expectation
- Independence

Joint PDF

- Two continuous random variables associated with the same experiment are jointly continuous and can be described in terms of a joint PDF $f_{X,Y}(x,y)$ if $f_{X,Y}(x,y)$ is a non-negative function that satisfies

$$P((X,Y) \in B) = \iint_{(x,y)\in B} f_{X,Y}(x,y)\,dx\,dy$$

for every subset B of the two-dimensional plane

- The probability $P(a \leq X \leq b,\ c \leq Y \leq d)$ can be computed from the joint PDF as follows

$$P(a \leq X \leq b, c \leq Y \leq d) = \int_c^d \int_a^b f_{X,Y}(x,y)\,dx\,dy$$

Properties of joint PDF

- All $f_{X,Y}(x,y)$ must be non-negative

$$f_{X,Y}(x,y) \geq 0 \quad , \quad \forall (x,y)$$

- Sum over all possibilities must be 1

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x,y)\, dx\, dy = 1$$

Example A1

- The joint PDF of X,Y is given as

$$f_{X,Y}(x,y) = \begin{cases} 6e^{-3x}e^{-2y} & , \ x \geq 0, y \geq 0 \\ 0 & , \ \text{otherwise} \end{cases}$$

  - Find $P(0 \leq X \leq 1, 0 \leq Y \leq 2)$

Marginal PDF

- Marginal PDF of *X*

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy$$

- Marginal PDF of *Y*

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx$$

Example A2

- The joint PDF of X,Y is given as

$$f_{X,Y}(x,y) = \begin{cases} 6e^{-3x}e^{-2y} & , \ x \geq 0, y \geq 0 \\ 0 & , \ \text{otherwise} \end{cases}$$

  - Find marginal PDF of X

Joint CDF

- If X and Y are two random variables associated with the same experiment, their joint CDF can be given as

$$F_{X,Y}(x,y) = P(X \leq x, Y \leq y)$$

$$F_{X,Y}(x,y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f_{X,Y}(s,t)\, dt\, ds$$

- The joint PDF can be obtained from the CDF as follows

$$f_{X,Y}(x,y) = \frac{\partial^2 F_{X,Y}}{\partial x \partial y}(x,y)$$

Conditional PDF

- The conditional PDF of X given $Y = y$, is given as follows

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} \quad , \quad f_Y(y) > 0$$

Example B1

- The joint PDF of X,Y is given as

$$f_{X,Y}(x,y) = \begin{cases} \frac{12}{5}x(2-x-y) & , \ 0 < x < 1, 0 < y < 1 \\ 0 & , \ \text{otherwise} \end{cases}$$

    - Find $f_{X|Y}(x|y)$

Example from [Ross, 2010]

Conditional Expectation

- Conditional expectation of $X|Y = y$

$$E[X|Y = y] = \int_{-\infty}^{\infty} x f_{X|Y}(x|y)\, dx$$

- Finding E[X] from conditional expectation

$$E[X] = E[E[X|Y = y]] = \int_{-\infty}^{\infty} E[X|Y = y] f_Y(y)\, dy$$

Example B2

- The joint PDF of X,Y is given as

$$f_{X,Y}(x,y) = \begin{cases} \frac{12}{5}x(2-x-y) & , \ 0 < x < 1, 0 < y < 1 \\ 0 & , \ \text{otherwise} \end{cases}$$

  - Find E[X | $Y = 0$]

Independence of 2 RV

- If *X* and *Y* are independent continuous random variables, then the joint PDF is equal to the product of the marginal PDF

$$f_{X,Y}(x,y) = f_X(x)f_Y(y)$$

Example A3

- The joint PDF of X,Y is given as

$$f_{X,Y}(x,y) = \begin{cases} 6e^{-3x}e^{-2y} & , \; x \geq 0, y \geq 0 \\ 0 & , \; \text{otherwise} \end{cases}$$

  - Are X and Y independent?